Mapping Structural Change in Manufacturing: Introducing the STiM Database*

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Latest version of the paper | STiM Database repository

Abstract. This paper introduces the Structural Transformation in Manufacturing Database (STiM), which provides employment, value-added, and output data at the two-digit manufacturing sub-sectoral level. In its current version, the database covers 12 homogenised industrial activities across more than 145 countries from 1963 to 2019. It is the result of combining several data sources and undertaking substantial harmonisation efforts aimed at ensuring internal, intertemporal, and international consistency. In addition, the database aims in linking several price deflators from multiple institutions to adjust nominal measures, thereby producing estimates of real value-added and constant output. To date, this makes the STiM database the most comprehensive dataset for long-run analyses of structural change within manufacturing in both developed and developing economies. The final database and the replication package will be made freely available at the corresponding GitHub repository .

Keywords: Manufacturing, structural change, database, employment, value-added, output.

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1 Introduction

One of the earliest and perhaps most important tenets of the literature on economic convergence is that development entails structural change (Kuznets, 1973; Allen, 2011). In this setting, industrialisation has long been considered the cornerstone of this transformation, as the reallocation of production factors towards manufacturing has been associated with several direct and indirect gains (Prebisch, 1950; Lewis, 1954; Hirschman, 1958; Kaldor, 1966; Szirmai, 2012). Yet, recent empirical evidence discussing the structural transformation of some developing countries has reshaped the debate on the role of manufacturing in economic development. In the aftermath of the Second World War, the tertiary sector in these economies has often expanded ahead of a sustained industrial base, while manufacturing has begun to decline at much lower income levels than in the past — a pattern now termed premature deindustrialisation (Palma, 2005; Dasgupta and Singh, 2007; Rodrik, 2016; Tregenna, 2016). Put differently, the traditional hump-shaped relationship between income per capita and manufacturing appears to have shifted downward and leftward over time, leading to projected decreases in industrial employment and value-added shares at earlier stages of development. This new time-dependent pattern of structural change hence suggests that industries have become a more challenging route to growth than in the past years, providing developing countries with a narrower window to reap the gains associated with industrialisation.

Given that these findings entail striking normative implications for development policies, a branch of new studies has recently emerged seeking to substantiate, validate, or mitigate these conclusions. Different perspectives have been chosen to contribute directly or indirectly to this ongoing debate, including discussions on whether this pattern still holds when examining the trajectories of new developing countries using exclusive datasets (Kruse et al., 2022; Nguimkeu and Zeufack, 2024; Lautier, 2024) or through reassessments of the contribution of manufacturing to productivity or economic growth (Szirmai and Verspagen, 2015; Cantore et al., 2017; Diao, McMillan et al., 2019; Forero and Tena-Junguito, 2024). Alternatively, further contributions have examined the factors driving successful industrialisation (Haraguchi, Martorano et al., 2019) and gauged the effects of significant post-1990s institutional and technological changes that may have either reduced manufacturing's competitiveness or undermined its historical role as an engine of growth (Baldwin, 2011; Autor et al., 2013; Bogliaccini, 2013).

Yet, so far, the main recurrent limitation of these analyses has been the trade-off between the level of aggregation and the number of countries covered. On the one hand, papers often rely on aggregate measures of industrialisation and deindustrialisation (e.g., manufacturing's share in GDP or employment), which provide broad cross-country coverage but mask heterogeneity across sub-sectors (Felipe et al., 2019). On the other hand, others use granular or firm-level data that provide valuable insights into within-manufacturing dynamics but confine the analysis to a small set of countries, thereby limiting the external validity of the findings (Diao, Ellis et al., 2025). This trade-off has made it difficult to reach generalisable conclusions, as one cannot infer generic

properties of manufacturing from a few countries, nor assume that premature deindustrialisation is a universal feature of the sector. Instead, it may be a phenomenon confined to specific traditional activities that were more affected by the technological and institutional changes that have been ongoing since the 1990s. To address these limitations, a few studies have chosen to conduct their analysis using datasets aggregated at the manufacturing sub-sectoral level, as this approach uncovers within-manufacturing dynamics while covering a greater number of countries (Rodrik, 2013; Vaz and Baer, 2014; Andreoni and Tregenna, 2020; Dosi et al., 2021; Bekhti, 2025). Although this is likely the most appropriate level of aggregation for reassessing conclusions about manufacturing's role in recent years, only a few datasets were available until recently. In addition, these datasets were either methodologically weak or incomplete and lacked consistent panel structures, rendering them unsuitable for reliable analysis (Rodrik, 2013).

This paper addresses these methodological limitations by introducing the Structural Transformation in Manufacturing (STiM) Database. The STiM database synthesises, aggregates and harmonises data from major institutional organisations and renowned research centres. Its aim is to provide a comprehensive and comparable unbalanced panel dataset on employment, value added, and output at the two-digit manufacturing sub-sectoral level, covering a wide set of countries over the longest time period possible. Constant output and real value-added measures are also provided, as the database links several price deflators from multiple institutions to adjust nominal values and thus avoid conflating prices with quantity. In its current version, the database covers 12 homogenised industrial activities across more than 145 countries from 1963 to 2019. To date, this makes the STiM database the most comprehensive dataset for long-run analyses of structural change within manufacturing in both developed and developing economies.

Beyond this initial release, the code infrastructure is designed to facilitate the updating of major data sources from traditional institutions as new vintages become available. It also supports the seamless incorporation of additional country-specific databases. These will be progressively added to enhance the quality of the estimates, as national statistical offices provide them in response to some of our requests. In line with this flexible design, the database is released in two versions to accommodate different user needs. The *ready-to-use version* contains the five core variables discussed in the next section (employment, nominal value-added, real value-added, nominal output, and real output), along with a limited set of metadata that identify the country, the year, and the relevant manufacturing sub-sector. This version is accompanied by an additional *detailed version*, which contains several tag variables intended to improve the transparency and tractability of the harmonisation and estimation procedures. These tags allow users to trace the source of each observation, the estimation methods applied (if any), the reference year used in the chain-linking procedure, the final configuration number identifying the reporting methods, the source of the deflator, and the deflator itself. It also includes a set of additional variables that were not retained in the core dataset, such as a regional classification and an adapted OECD

industry classification capturing the technological intensity of each of the 12 sub-sectors (OECD, 2003; Vu et al., 2021). Finally, a replication package is provided to ensure full reproducibility and to allow users to adjust any methodological assumptions applied in the estimation procedure, should these be deemed too conservative or too aggressive. The final database (V1) and replication files will be made freely available through the corresponding GitHub repository[1].

The construction of the STiM database involved several steps. First, our methodological approach began by retrieving the latest version of the INDSTAT database from UNIDO (Revision 3), which serves as our baseline since it provides the longest temporal and broadest country coverage (UNIDO, 2024). We then downloaded external datasets from major institutions containing data at the manufacturing sub-sector level. After defining common boundaries for the manufacturing sector and extracting only employment, value-added, and output values from these providers, we rescaled units and harmonised currencies.

Second, even when expressed in the same unit, these sources cannot simply be combined as they often do not rely on the same sample, industrial classifications, or reporting methods. In addition, INDSTAT raw values — which serve as the baseline for every final series — contain several temporal inconsistencies that may be overlooked when downloading directly consolidated variables from UNIDO. As such, every cross-country or intra-country analysis relying on these raw data is likely to be erroneous. To name but a few, national statistical offices may change their industrial classification over time, leading to spurious shifts in the sectoral composition of manufacturing. For instance, they might report values for the textile industry while also including apparel in some years, or report food products while also including beverages and tobacco in other years. In its native form, INDSTAT reports more than 134 different possible combinations of activities at the two-digit level, which vary across countries, variables, and years. Another issue relates to reporting methods that have evolved, following changes in national accounting standards (United Nations, 1968b; International Monetary Fund, 2025). Each national statistical office may change the way employment, value-added, or output is reported over time, leading to unintended breaks in the series. While employment can be reported as the number of employees or number of persons engaged, value-added and output can be reported at basic prices, producer prices, factor values, or unknown classification with potentially several changes occurring within the same country. These two issues (among others) make it impossible to rely directly on INDSTAT or to combine it with external sources without undertaking substantial aggregation and harmonisation efforts. To address these challenges, the second step consisted of developing a systematic and transparent harmonisation procedure, relying on chain-linking techniques and a set of conservative assumptions to ensure internal, intertemporal, and international consistency.

[1]The latest version of the STiM database and the replication package can be publicly accessed through the following link: https://github.com/yanisbkt-econ/STIM-Database .

Finally, the last step of the construction process involved linking several price deflators from multiple institutions to adjust nominal measures, thereby producing estimates of real value-added and constant output. This step was particularly challenging, as deflators are often not available at the two-digit manufacturing sub-sector level for a wide set of countries, particularly developing economies, and over a long period. To mitigate this issue, we first draw on Haraguchi and Amann (2023), who derive sub-sectoral price indices from the Index of Industrial Production (IIP) embedded in INDSTAT. Since the IIP is designed to measure real production growth for each manufacturing activity and thus captures volume changes relative to a baseline year (2015), it indirectly provides price indices that can be retrieved and used to deflate nominal value-added. To complement this deflator, we intend to further retrieve price indices from external sources (on going). So far, on the absence of other solution, we rely on the GDP deflator harmonised by Müller et al. (2025) to provide estimates on the constant value-added and output when there is no deflator at the sub-sectoral level.

This paper contributes to the existing literature by extending previous academic efforts to compile and harmonise macroeconomic statistics for a wide range of countries. To our knowledge, very few studies have attempted to build a comprehensive panel at the manufacturing sub-sector level. This is primarily because INDSTAT — the most comprehensive source of information covering manufacturing sub-sectors — has only been freely available since February 2022. So far, the closest attempt to build such a panel was undertaken by Pahl and M. P. Timmer (2020). Although their primary focus was not to provide a harmonised database, they nonetheless constructed a panel combining the same data sources as this paper, relying on the same kind of chain-linking procedure. Yet, we believe this paper substantially improves on their pioneering work in several ways. First, we refine the cleaning procedure to detect spurious zeros and outliers in the raw INDSTAT observations. Second, our harmonisation procedure extends coverage back to the 1960s and up to 2019, thereby lengthening the period covered by more than twenty years. To achieve this, we enhance the chain-linking method to maximise the continuity of each country's series by allowing for several possible reference years. Third, our database is fully reproducible and will be further extended as new sources become available in response to our requests. Lastly, we aimed in providing real value-added and constant output measures by linking several price deflators from multiple institutions to adjust nominal measures. This makes the STiM database the most comprehensive dataset for long-run analyses of structural change within manufacturing in both developed and developing economies.

The paper proceeds as follows. Section 2 presents the main manufacturing data sources and discusses the challenges involved in harmonising and combining them. Section 3 discusses the data cleaning procedures applied to each dataset, and Section 4 presents the industrial classification adopted in the STiM database. This results in an aggregation into 12 sub-sectors, hereafter defined as the 'STiM classification'. While this approach reduces cross-sector variation, it addresses the issue of unusually reported combined activities and facilitates linkages with external datasets. Section 5 briefly discusses linear interpolation and its limitations, while Section 5 details the chain-linking procedure

used to ensure internal, intertemporal, and international consistency. Several new techniques are introduced to maximise the continuity of each country's series. Section 7 describes the price indices and the deflation procedure, while Section 8 presents a few descriptive statistics for the STiM database. Finally, Section 9 concludes and discusses potential extensions of the database.

2 Presentation of the main data sources

This first section aims to present the main data sources used to construct the Structural Transformation in Manufacturing (STiM) Database. Consequently, we begin by discussing the INDSTAT Revision 3 database, which serves as the baseline source of information for employment, value added, and output at the two-digit manufacturing sub-sector level. Subsequently, we describe the remaining public datasets that are linked to INDSTAT in order to complement its coverage. We ultimately connect more than ten external databases originating both from major institutional organisations and renowned research centres. Lastly, we conclude this section by summarising the main challenges that arise when trying to perform such linkages across these datasets.

2.1 The INDSTAT Revision 3 Database

The main source for the Structural Transformation in Manufacturing (STiM) database is INDSTAT, which compiles national industrial surveys and representative censuses collected by the United Nations Industrial Development Organisation (UNIDO, 2024). These surveys exclude informal firms covering only registered activities with more than five and sometimes ten employees, depending on the censuses. This extensive data collection, conducted since the 1960s and made freely available in February 2022, provides a wide range of variables at the two-, three-, and four-digit levels of aggregation according to the International Standard Industrial Classification of All Economic Activities (ISIC), Revision 3.1 (United Nations, 2002). For the purpose of this database, we focus exclusively on employment, nominal value added, and nominal output at the two-digit level of aggregation, which allows us to construct the longest possible time series. It should be noted that all monetary variables are initially retrieved in local currency units (LCU), although UNIDO also provides series expressed in current US dollars using period-average bilateral exchange rates. Yet, we instead depart from LCU values to avoid some inconsistencies found in the implicit exchange rate used by UNIDO, as discussed in Section 3.

The initial raw data cover 186 countries and 31 industrial sub-sectors, spanning the period from 1963 to 2022. Yet, this panel is unbalanced, with many gaps across years, sub-sectors, and variables, so that each country may enter or exit the sample several times during the period. In order to ensure consistency and comparability across all datasets, we first limit the sample to 2019 to avoid the decline in observations due to ongoing reporting. Secondly, we restrict the sub-sectors to manufacturing activities only,

corresponding to those whose ISIC codes range from 15 to 37 (United Nations, 2002). The details of these 23 manufacturing sub-sectors are provided in Table **??**. Overall, this very preliminary cleaning step results in a raw sample of 23 manufacturing sub-sectors across 179 countries, with 109,246 observations for employment, 103,837 for output, and 98,230 for value added.

Table 1: ISIC Rev. 3.1 Manufacturing Categories (Section D)

ISIC, Rev. 3.1	**Initial ISIC Categories available**
15	Food and beverages
16	Tobacco products
17	Textiles
18	Wearing apparel, fur
19	Leather, leather products and footwear
20	Wood products (excl. furniture)
21	Paper and paper products
22	Printing and publishing
23	Coke, refined petroleum products, nuclear fuel
24	Chemicals and chemical products
25	Rubber and plastics products
26	Non-metallic mineral products
27	Basic metals
28	Fabricated metal products
29	Machinery and equipment n.e.c.
30	Office, accounting and computing machinery
31	Electrical machinery and apparatus
32	Radio, television and communication equipment
33	Medical, precision and optical instruments
34	Motor vehicles, trailers, semi-trailers
35	Other transport equipment
36	Furniture; manufacturing n.e.c.
37	Recycling

These three main consolidated variables cannot yet be fully exploited in their current form, as they are merely a combination of the different reporting methods used to measure employment, value added, and output. They indirectly reflect changes in the System of National Accounts (SNA) recommendations, which have evolved over time (United Nations, 1968b; International Monetary Fund, 2025). Importantly, these changes are specific to each country and year, with absolutely no common pattern over time or across regions, as national statistical offices may or may not apply these rules. As such, these three initial consolidated variables are not suitable for empirical analysis, since each change in reporting methods introduces unintended variations, notably level shifts, in the series. As a result, we discard these consolidated series and recover the initial ten unconsolidated variables that correspond to each reporting method (cf. Table 2). Two variables correspond to employment, measured either as the number of employees or as the number of persons engaged. The remaining eight variables correspond to value added and output, each reported under three different valuation methods (basic prices, producer prices, and factor cost), plus an additional category of unknown classification.

Three points warrant particular attention. First, for a given country-year observation, statistical offices report each variable under only one definition or valuation method. For example, employment is reported either as the number of employees or as the number of persons engaged, but never both simultaneously. Similarly, value added and output are reported under only one valuation method at a time. Second, value added and output variables, when available, are not necessarily reported using the same valuation method for a given country-year observation, which introduces further inconsistencies. Third, one should not underestimate the bias each change in reporting methods may introduce.

For example, in Brazil, the statistical office changed the definition of employment in 1996, switching from the number of employees to the number of persons engaged. This led to a sudden increase of hundreds of thousands of workers in activities with a high prevalence of self-employment, such as in the Food and Beverages industry (ISIC 15, Rev. 3.1). Indeed, the reported value nearly doubled, moving from 596,406 employees in 1995 to 1,008,577 persons engaged in 1996. Although less pronounced than in the case of employment measurement, a similar level shift may exist across the three valuation methods for reporting value added and output. In a nutshell, under basic price valuation, subsidies on products are included while taxes are excluded to reflect what the producer actually receives (International Monetary Fund, 2025, p.214). Conversely, under producers' prices, taxes are included and subsidies excluded to assess what the producer charges (International Monetary Fund, 2025, p.214). Lastly, under factor cost, both taxes and subsidies on products are excluded, and value added is obtained by adjusting value added at basic prices for other taxes and subsidies on production (International Monetary Fund, 2025, p.218). Given that taxes and, notably, subsidies are well-known instruments of industrial policy (U. C. V. Haley and G. T. Haley, 2013; Aiginger and Rodrik, 2020), these different valuations methods can indeed lead to discrepancies, depending on each country, year, and sub-sector. The details on the frequency of these ten unconsolidated variables are provided in Table 2

Another concern with this dataset relates to how the raw 23 sub-sectors are defined in the ISIC Rev. 3.1 classification and how this structure is maintained across earlier ISIC revisions. At first glance, the dataset logically imposes a panel structure that includes 23 manufacturing sub-sectors, consistent with the ISIC Rev. 3.1 classification at the time of release. Yet, a closer look reveals that some sub-sectors only began to be reported after a certain year, which often coincides with the introduction of new ISIC revisions. In earlier decades, previous ISIC revisions were in force, and certain industrial activities that are now reported separately were then aggregated into broader categories, reflecting differences in classification norms. For instance, manufacturing activities linked to leather products and the footwear industry (ISIC 19, Rev. 3.1) were only distinguished from the wearing apparel industry (ISIC 18, Rev. 3.1) in the early 1990s. This became possible only with the introduction of the ISIC Rev. 3 classification in 1989 (United Nations, 1989). Consequently, there are no historical data for these activities prior to this date since national statistical institutes were all following the ISIC Rev. 2 classification then in force (United Nations, 1968a).

Table 2: Summary statistics of the unconsolidated variables

Variables	N	Mean	Min	Max
Employment				
Employees	93,142	94,508.51	0	10,200,000
Persons engaged	16,104	44,519.07	0	1,871,000
Value added				
Basic prices	38,877	7.93e+08	-6.12e+08	1.85e+11
Producer prices	30,272	3.05e+09	-1.28e+09	5.90e+11
Factor cost	11,252	3.05e+09	-8.29e+08	1.39e+11
Unknown classification	17,829	7.39e+09	-4.89e+09	5.53e+11
Output				
Basic prices	9,754	5.63e+09	0	2.53e+11
Producer prices	45,282	3.76e+09	0	1.30e+12
Factor cost	17,023	5.51e+09	0	5.04e+11
Unknown classification	31,778	1.88e+10	0	2.05e+12

Unfortunately, these issues of industry combination are not confined to such clear-cut and rational cases. There are many more complex cases where countries report certain sub-sectors jointly for no apparent reason, even though the industrial classification in force at the time does not recommend it. In its raw form, INDSTAT reports more than 134 different possible combinations of activities at the two-digit level, which vary across countries, variables, and years. To give a concrete example, Peru reports correctly employment as the number of persons engaged in the Food and Beverages sector (ISIC 15, Rev. 3.1) without any combination from 1979 (the year of its entry into the database) to 2003. Then, for whatever reason, from 2004 to 2013 it reports the Food and Beverages sector (ISIC 15, Rev. 3.1) combined with Tobacco products (ISIC 16, Rev. 3.1), before reverting to the original classification from 2014 to 2019. It is also in 2014 that Peru began reporting employment as the number of employees instead of persons engaged, thereby adding further noise and united variation to the series. As such, relying directly on the original consolidated variables provided by INDSTAT is likely to lead to erroneous conclusions as it mixes different valuations methods but also because the composition of each sub-sector might changes over time.

Overall, these issues of comparability and consistency across countries, variables, and over time must be addressed before any meaningful analysis can be conducted — the solution proposed to mitigate these issues are discussed in section 6.

2.2 External disaggregated datasets used to enhance estimates and coverage

To enhance the coverage of the final database, we aimed to link all available datasets on employment, value added, or output at the two-digit manufacturing subsector level — what we refer to as 'disaggregated' databases. To our knowledge, only six main public datasets meet these criteria regarding employment, and seven regarding value added

and output, resulting in the integration of nine unique datasets. They are all presented in detail below. Each of these datasets was integrated in its latest available version, i.e., released in 2025 at the time of writing this draft, and will be updated as new vintages come out.

2.2.1 The OECD STAN Database

The STructural ANalysis (STAN) industry database is built on national accounts and industrial surveys harmonised by the OECD (Horvát and Webb, 2020). It provides internationally comparable series on output, value added, and employment at the 2- and 3-digit levels, consistent with the System of National Accounts 2008 (International Monetary Fund, 2009). This implies that every two-digit subsector (which we retain[2]) is defined uniformly across countries and over time, thereby avoiding spurious variations arising from differences in the composition of activities within subsectors. In most cases, employment is reported both as the number of employees and as the number of persons engaged, while value added is reported at basic prices and at factor cost. However, output is always reported under one valuation method, namely basic prices.

Unlike INDSTAT, a single country-year-sector observation can thus be reported in multiple ways simultaneously for a given variable (output, employment, or value added). Yet, in practice, there are no cases where value added is reported at factor cost but not at basic prices. We therefore discard the series reporting value added at factor cost, as it does not improve coverage and is one of the valuation methods least privileged in the chain-linking procedure (see Section 6). However, we retain both valuation methods for employment, as some countries report only one of the two definitions. This is especially relevant for early years when data regarding employment status were scarcer (e.g., Canada, France). We ultimately end up with four main series stemming from the OECD STAN database, namely, employment expressed as the number of employees, employment expressed as the number of persons engaged, value added at basic prices, and output at basic prices. In these last two cases, all monetary values are expressed in local currency units (LCU) and reported in nominal terms. Although real measures are available, we rely on the undeflated data to ensure consistency with INDSTAT and subsequently deflate the final STiM series in a uniform manner, as explained in Section 7.

Regarding coverage, the series generally begin in the early 1970s and extend to the most recent year available, which we restrict to 2019 to ensure consistency with the other datasets. As implied by its name, the database covers the 38 OECD members, although with varying degrees of exhaustiveness. While a country cannot appear and disappear over time, the panel remains unbalanced, as not all country series start in the same year. For example, the series for France start in 1970, those for Portugal in 1980, those for Australia in 1990, and those for Colombia in 2000. Although the panel tends to be biased towards high-income countries, it also includes several emerging

[2]The STiM Database is constructed at the two-digit level; therefore, we discard every observation at a finer level of aggregation.

economies, such as Chile, Mexico, Costa Rica, and Türkiye, which makes it a valuable complement to INDSTAT — the baseline dataset for STiM.

Lastly, in terms of industrial classification, the OECD STAN database follows the NACE Rev. 2 classification, which fully corresponds to the ISIC Rev. 4 classification (European Commission, 2008; United Nations, 2008). More details on the 24 included sub-sectors can be found in Table 3. Yet, for some countries and sub-sectors, the OECD STAN database does not report data at the individual sub-sector level but rather for aggregated combinations. While such cases of aggregation remain limited, the database consistently reports, regardless of the country, data for the combined activities of the manufacture of furniture (ISIC 31, Rev. 4) and other manufacturing activities (ISIC 32, Rev. 4). This prevents the retrieval of separate values for these two activities, making it the only systematic case in which neither sub-sector is reported individually. The exact distribution by ISIC Rev. 4 manufacturing codes in the OECD STAN database is provided in Table A1.

Table 3: ISIC Rev. 4 Manufacturing Categories (Section C)

ISIC, Rev. 4	**Manufacturing Categories**
10	Manufacture of food products
11	Manufacture of beverages
12	Manufacture of tobacco products
13	Manufacture of textiles
14	Manufacture of wearing apparel
15	Manufacture of leather and related products
16	Manufacture of wood, etc.
17	Manufacture of paper and paper products
18	Printing and reproduction of recorded media
19	Manufacture of coke and refined petroleum products
20	Manufacture of chemicals and chemical products
21	Manufacture of basic pharmaceutical products, etc.
22	Manufacture of rubber and plastics products
23	Manufacture of other non-metallic mineral products
24	Manufacture of basic metals
25	Manufacture of fabricated metal products, etc.
26	Manufacture of computer, electronic and optical products
27	Manufacture of electrical equipment
28	Manufacture of machinery and equipment n.e.c.
29	Manufacture of motor vehicles, trailers and semi-trailers
30	Manufacture of other transport equipment
31	Manufacture of furniture
32	Other manufacturing
33	Repair and installation of machinery and equipment

2.2.2 The OECD TiVA Database

The Trade in Value Added (TiVA) database by the OECD is a collection of measures providing insights into global production networks and supply chains, covering 45

sectors up to the four-digit level (OECD, 2022; OECD, 2025). While this input-output table usually serves to track the origins of value added in exports, imports, and final demand, it also includes data on value added and output at a highly disaggregated level across all sectors. Both variables are reported under a single valuation method, namely basic prices, and are expressed in current US dollars. Although it does not cover as many years as the OECD STAN database (as it only starts in 1995), the TiVA database covers many more countries, as it is directly derived from the OECD Inter-Country Input-Output (ICIO) tables. Indicators are provided for 80 economies, including all OECD, European Union, ASEAN, and G20 countries, thereby covering many emerging economies and making it a valuable complement. This is especially true in the most recent version (i.e., 2025), in which some new African countries are now covered, such as Angola (AGO), São Tomé and Príncipe (STP), and the Democratic Republic of the Congo (COD). As with the STAN database, the TiVA database complies with the 2008 SNA standards (International Monetary Fund, 2009) and follows the ISIC Rev. 4 classification depicted in Table 3. To make it consistent with previous databases, we restrict the years to 2019.

Regarding the coverage of sectors, we restrict the TiVA database to activities belonging exclusively to manufacturing (i.e., from C10 to C33) as defined by the ISIC Rev. 4 classification. However, similar to the OECD STAN database, some sub-sectors are not reported individually but rather in combined forms. While this tends to happen more frequently than in the STAN database, it provides a perfectly balanced database regarding both value added and output for all the manufacturing sub-sectors reported. The details of the frequency of these ISIC Rev. 4 codes are provided in Table A2.

2.2.3 The OECD TiM Database

Designed to complement the TiVA database, the OECD Trade in Manufacturing (TiM) dataset is a collection of labour market indicators aimed at providing additional insights into global production networks and supply chains (Horvát, Webb and Yamano, 2020). Still relying on the OECD's Inter-Country Input-Output (ICIO) Tables, it also covers 45 individual sectors based on the ISIC Rev. 4 classification and complies with 2008 SNA standards. While most variables report employment by type of production network (i.e., domestic employment embodied in exports, etc.), it also includes a simple measure of employment by industry. In this series, employment is only reported in one way, namely as persons engaged in production activities, thus including both employees and the self-employed. The preferred unit of measurement is persons although, if such statistics are unavailable, then the number of jobs by industry is used (e.g., Israel, Mexico, the US, etc.). As stated in the documentation, estimates from National Accounts statistics are always preferred over Labour Force Surveys (LFS) since the latter usually exclude non-resident workers while including resident workers commuting abroad. More details on each country case can be found in the original documentation (Horvát, Webb and Yamano, 2020, p.43).

Regarding the database coverage, it covers 61 countries from 1995 to 2020, which we restrict to 2019 for consistency. Similar to the TiVA database, it thus includes many emerging economies, yet with a somewhat reduced sample in comparison with the former. Lastly, the sub-sector structure follows exactly the same classification as OECD TiVA, implying that, once more, some sub-sectors are not reported individually but are rather combined. Details are provided in Table A3.

2.2.4 The EU KLEMS Database

The EU KLEMS database provides detailed employment, value added, and output data across 40 industries up to the two-digit level and using the ISIC Rev. 4 classification (Bontadini et al., 2023). It complies with the 2008 SNA standards, with value added and output therefore reported (only) at basic prices and employment being reported both as the number of employees and as the number of persons engaged. The coverage of employment series overlaps perfectly; therefore, we discard the latter and retain only employment as the number of employees. Regarding monetary values, they are all expressed in local currency units (LCU), and we only keep series reported in nominal terms. Although real measures are available (just like with OECD STAN), we rely on the undeflated data to ensure consistency with INDSTAT and subsequently deflate the final STiM series in a uniform manner, as explained in Section 7.

Regarding the coverage, the database covers exactly 30 countries, namely the 27 EU Member States, the United States, Japan, and the United Kingdom (Bontadini et al., 2023). While the coverage is thus more limited than in previous external databases, it still includes some emerging economies, especially those belonging to the EU but not part of the OECD. This concerns countries such as Bulgaria, Croatia, Cyprus, Malta, and Romania. Series span the period 1995-2021, albeit with some gaps and missing values, but are restricted to 2019 for consistency. Regarding sub-sector coverage, it exhibits frequent combinations of activities across sub-sectors. While the OECD STAN database fails to report only two out of 24 sub-sectors individually (in a systematic way), the EU KLEMS database does not report up to 18 out of 24 sub-sectors individually. Most of them are systematically retrieved in a combined form across all countries and years. For example, the manufacture of food products (ISIC 10, Rev. 4) is always reported jointly with the beverage industry (ISIC 11, Rev. 4) and the manufacture of tobacco products (ISIC 12, Rev. 4). The only sub-sectors that are consistently reported individually are the manufacture of coke and refined petroleum products (ISIC 19, Rev. 4), the manufacture of chemicals (ISIC 20, Rev. 4), the manufacture of basic pharmaceutical products (ISIC 21, Rev. 4), the manufacture of computer, electronic, and optical products (ISIC 26, Rev. 4), the manufacture of electrical equipment (ISIC 27, Rev. 4), and the manufacture of machinery and equipment (ISIC 28, Rev. 4). More details are provided in Table A4.

2.2.5 The CEPAL BADECON Database

The CEPAL BADECON database is compiled by the Economic Commission for Latin America and the Caribbean (ECLAC) and provides data regarding manufacturing employment, value added, and output. While most of the series are aggregated at the country–year level, some disaggregated observations are also available for a limited number of countries and years. We thus exploit this somewhat limited coverage to complement our database. Note that the technical sheet provided along with each series is rather limited, so we do not know much about the precise definitions used to measure employment, value added, or output. We simply know that all monetary variables are expressed in current US dollars and that it follows the ISIC Rev. 4 classification.

Regarding the coverage, disaggregated series are only available for 9 countries, namely Brazil, Chile, Colombia, Ecuador, El Salvador, Mexico, Panama, Peru, and Uruguay. They span different periods depending on the country, with the earliest series starting in 2010 and extending until 2021, which we restrict to 2019 for consistency. Regarding the sub-sector coverage, there is no combination of activities, meaning that all 23 manufacturing sub-sectors are reported individually, yet with missing values in some cases (e.g., no value added and employment for Peru, etc.). More details are available in Table A5.

2.2.6 The GGDC Input Output Database (WIOD)

The World Input-Output Database (WIOD) from the Groningen Growth and Development Centre (GGDC) provides harmonised input-output tables for 28 EU countries and 15 other major economies over the period 2000–2014 (M. P. Timmer et al., 2015). While these tables are mostly used to study bilateral trade flows between sectors and countries, we can also derive from these flows the value added and output produced by each sector domestically — mirroring the process described for the OECD TiVA database. In its 2016 version, it becomes consistent with the 2008 System of National Accounts, implying that both value added and output are reported at basic prices, with values being expressed in current US dollars. While the coverage regarding the number of countries and years appears to be quite limited in comparison with previous datasets, we still include it as a backup source, given that the panel is perfectly balanced with no missing values. Regarding the sub-sector structure, it follows the ISIC Rev. 4 classification depicted in Table 3. Moreover, only three systematic combinations of activities exist across all countries and years, namely the manufacture of food (ISIC 10, Rev. 4) combined with beverages (ISIC 11, Rev. 4) and tobacco products (ISIC 12, Rev. 4), as well as the manufacture of textiles (ISIC 13, Rev. 4) combined with wearing apparel (ISIC 14, Rev. 4) and leather and related products (ISIC 15, Rev. 4). Lastly, just like most previous databases, the manufacture of furniture (ISIC 31, Rev. 4) and other manufacturing (ISIC 32, Rev. 4) are always reported jointly. More details are provided in Table A6.

2.2.7 The GGDC World Input-Output Database (H-WIOD)

The Historical World Input–Output Database (WIOD) provides annual time series of world input–output tables spanning the period 1965-2000 (Woltjer et al., 2021).. This long-run version of the WIOD database, also maintained by the Groningen Growth and Development Centre (GGDC), allows for a much larger time span, serving as a secondary source for pre-1970 years other than INDSTAT. It is consistent with the 1993 System of National Accounts, with both value added and output reported at producer prices, with values being expressed in current US dollars. As stated by the documentation, the database is constructed using a top-down reconciliation approach that combines national accounts, benchmark input–output tables, and bilateral trade statistics, with missing years imputed through constrained matrix balancing procedures (Woltjer et al., 2021).

Regarding the coverage, only 25 countries are included in the database, mostly European ones, with the exception of Australia, Brazil, Canada, China, Hong Kong, India, Japan, Korea, Mexico, Taiwan, and the USA. Together, these countries account for roughly 85% of world GDP over the period 1965-2000. The March 2022 release follows the ISIC Rev. 3.1, just like the INDSTAT classification, with a balanced panel structure. It adopts a 12 manufacturing sub-sector structure: six individual sub-sectors, five groups of industries that are combined and one residual category. Yet all manufacturing codes are harmonised across countries and years, with missing observations extrapolated using growth-consistent shares from auxiliary sources. More details on these codes are provided in Table A7.

2.2.8 The APO Database

The APO Productivity Database 2025, documented in the APO Productivity Databook 2025 (Nomura, 2025), provides a harmonised set of long-run productivity and growth accounts for Asian economies over the period 1970–2023. The database is constructed from official national accounts, population statistics, and labour force surveys reconciled within a unified growth accounting framework that aligns with the 2008 System of National Accounts. While the database primarily focuses on total economy-level aggregates, it also includes sectoral data for manufacturing industries at the two-digit level. Yet, only employment and value added can be retrieved in this database, with employment reported as the number of persons engaged and value added at basic prices. All monetary values are available in different units, but we retain only the LCU values expressed in nominal terms.

Regarding the coverage, the database covers in theory 31 Asian economies, including both high-income countries such as Japan and South Korea, as well as emerging economies like India, Indonesia, and Vietnam, and several island economies such as Fiji and Sri Lanka. The series span the period from 1970-2023, which we restrict to 2019 for consistency. Regarding the sub-sector structure, it follows its own aggregation scheme, yet a correspondence table is available that makes it possible to retrieve the

ISIC Rev. 4 classification depicted in Table 3. Quite many aggregate groups are reported in the database, with only four individual sub-sectors available (C16, C23, C24, and C27) that coexist within an unbalanced panel with substantial coverage discrepancies among different countries. For example, no series at the two-digit level are available for Myanmar, Nepal or Oman, while there are only full value-added series for Philipine (no employment) over the entire period. For example, no series at the two-digit level are available for Myanmar, Nepal, or Oman, while there are only full value-added series for the Philippines (no employment) over the entire period. More details regarding the distribution of non-missing observations across ISIC codes for each variable are provided in Table A8.

2.3 External aggregated datasets used to enhance estimates and coverage

While the main objective of the STiM database is to construct harmonised time series at the two-digit manufacturing sub-sector level, we also rely on a limited set of country-year manufacturing series covering total manufacturing employment, value added, and output — what we refer to as the 'aggregated' database. These series are used to approximate the growth rates of missing sub-sector series only when two disaggregated data streaks are observed on both sides of a temporal gap and when no other estimation technique can be applied. The rationale behind this choice, rather than accepting such interruptions (and allowing a country to disappear and reappear over time), is that the chain-linking procedure — required to ensure SNA compliance — relies on continuous growth rates with a fixed reference year. As such, having unconnected growth rate series would lead to the loss of the second segment separated by missing observations, and treating separated segments as independent would violate the internal and intertemporal consistency ensured by the chain-linking procedure. Further details on the methodology are provided in Section 6. This section simply presents the four aggregated datasets used to complement the disaggregated sources.

2.3.1 The GGDC-10 Sector Database

The GGDC-10 Sector Database (GGDC-10SD) is a collection of long-run sectoral accounts for 42 countries from the Groningen Growth and Development Centre (M. Timmer et al., 2015). It provides harmonised data on value added and employment for ten broad sectors (e.g., agriculture, construction, electricity, etc.) following the ISIC Rev. 3.1 classification, including a sector dedicated to manufacturing (ISIC D following Table 1). The database is constructed from official national accounts, population censuses, labour force surveys and establishment surveys, which are reconciled through a systematic linking and benchmarking procedure. Their methodology ensures consistency with SNA recommendations, with annual value added mostly reported at basic prices and employment as the number of persons engaged. No output series are available. Value added is expressed in local currency units (LCU) and reported both in nominal and real

terms. As with previous datasets, we retain only the nominal serie and subsequently deflate the final STiM series in a uniform manner, as explained in Section 7.

Regarding the coverage, the database covers 11 countries in Africa, 11 countries in Asia, 2 countries in the Middle East and North Africa (MENA), and 9 in Latin-America. For comparison, some data have been included for 8 European countries and for the United States. Series generally spans the period 1950-2012 with some variation in the starting periods especially for African countries. Otherwise, the panel is balanced with some estimation techniques applied to fill some small gaps. We restrict to the manufacturing sector discarding all other sectors.

2.3.2 The ETD and ASD Databases

The Economic Transformation Database (ETD) is constructed following the same logic as the GGDC-10SD, extending the time-span coverage to more recent years while initiating the series in 1990 (Kruse et al., 2022). It also increases the number of broader sectors covered from 10 to 12 (e.g., agriculture, construction, electricity, etc.) and makes it consistent with the ISIC 4 classification. As usual, we are only interested in retrieving data from the aggregated manufacturing sector (ISIC C). As before, value added is reported at basic prices and employment as the number of persons engaged, with all value added series expressed in local currency units (LCU) and reported both in nominal and real terms. We retain only the nominal series and subsequently deflate the final STiM series in a uniform manner, as explained in Section 7.

Regarding the coverage, it includes 20 countries in Asia (14 considered as developing Asia and 6 as high-income Asian countries), 9 countries in Latin America, 4 countries in the Middle East and 18 countries in Sub-Saharan Africa. This surge in African country coverage compared to the last release is possible with the inclusion of the African Sector Database (ASD) — a project that had been developed between the release of the GGDC and the ETD (G. d. Vries et al., 2015). All series span the period 1990–2018 with no missing values over the period. It no longer covers any European country nor the United States, which is not an issue as these countries are already those with the best coverage when combining all disaggregated datasets.

One drawback of the ETD database regarding our work is its shorter historical depth since its restriction to the 1990s. We thus rely on the linked series provided by the GGDC, which is a simple combination of the ETD database for recent years with the GGDC-10SD and the ASD for earlier periods, using backward extrapolation through growth rates of the respective series. This allows us to retrieve long-term series anchored to the more recent reliable values from the ETD database.

2.3.3 The ETD-TE Database

The Economic Transformation Database of Transition Economies (ETD-TE) is an extension of the ETD database specifically designed to include fourteen former-Soviet

Republics (Hamilton and G. J. d. Vries, 2025). The methodology aligned with the Economic Transformation Database, covering 12 broad sectors including manufacturing. As with the ETD database, value added is reported at basic prices and employment as the number of persons engaged, with all value added series expressed in local currency units (LCU) and reported both in nominal and real terms. Once more, we retain only the nominal series. The period covered spans from 1995 to 2019. Countries included are Armenia, Azerbaijan, Belarus, Estonia, Georgia, Kazakhstan, Kyrgyzstan, Latvia, Lithuania, Moldova, Russian Federadation, Tajikistan, Ukraine, and Uzbekistan.

2.3.4 The WDI Database (World Bank)

Lastly, we also rely on the World Development Indicators as a last resort to retrieve data on manufacturing value added at the country-year level (World Bank, 2022). No employment nor output series are available in this database, but we retain the value added series reported in current LCU. The coverage of countries is extensive, spanning the period from 1960 to 2024, which we restrict to 2019 for consistency. Yet, the starting point of the series varies greatly across countries, with some starting only in the early 2000s. Moreover, depending on the country, the way value added is reported might differ and usually mixes different valuation methods, which is why we rely on data points from this database only when no other source is available.

2.4 Challenges when linking all these external datasets to INDSTAT

Two main issues arise when linking these aforementioned datasets to INDSTAT, the baseline for the STiM database. First, this requires the construction of a correspondence table between the different classifications used across datasets to ensure meaningful sub-sector comparisons over time and across countries. While INDSTAT and the GGDC Historical WIOD database follow the ISIC Rev. 3.1 classification, most external disaggregated datasets rely on ISIC Rev. 4 classifications, as they generally cover more recent periods. In this exercise, a further complication arises from the fact that some external datasets report certain sub-sectors in combined forms rather than individually, as explained in the previous sections and described in different tables in the Appendix (see Tables A1–A8). This is also the case for INDSTAT, which contains 134 different combinations of activities. These constraints substantially limit the flexibility of any concordance exercise, particularly because the analysis cannot be conducted at a level more detailed than the two-digit level. Taken together, these limitations prevent the construction of a perfect correspondence table between the two classifications, as many manufacturing activities at a finer level are reallocated across different two-digit sub-sectors. Despite the impossibility of having a one-to-one mapping, Section 4 introduce a correspondence table that aggregates the original 23 (Rev. 3.1) or 24 (Rev. 4) manufacturing sub-sectors into 12 harmonised activities. We believe that this aggregation scheme (hereafter referred to as the STiM classification) is sufficiently detailed to preserve meaningful variation within manufacturing, while remaining

aggregated enough to limit data losses from combined activities and to minimize inconsistencies between ISIC Rev. 3.1 and ISIC Rev. 4. More details are provided in Section 4.

Second, even with a correspondence table, the question remains as to how to link these datasets to INDSTAT without distorting the resulting time series. Indeed, baseline series cannot be simply augmented or extended by substituting missing observations with values drawn from external sources. This is because the values within these datasets typically differ in their levels due to differences in valuation methods or sampling procedures (e.g., surveys versus national accounts, etc.). Critically, this concern also applies to INDSTAT itself, given that series are reported using different valuation methods as discussed in Section 2.1. Consequently, any change in the source or valuation method over time would introduce spurious level shifts in the final consolidated series. To mitigate this issue and ensure consistent series, we therefore rely on a chain-linking procedure that sets a common reference year stemming from the baseline series for each country. This procedure then relies on growth rates extrapolation from external datasets to reconstruct missing observations relative to this baseline year. Further details on this procedure are provided in Section 6, while Table 4 summarises the main characteristics of each database used to construct the STiM series.

3 Data cleaning, exchange rates and specific country-cases

This section presents all the changes made to the raw series prior to the aggregation into the 12 homogenised manufacturing sub-sectors and the chain-linking strategy. It first details the pre-cleaning procedure applied to the INDSTAT Revision 3 database, which is the most prone to inconsistencies. It then discusses specific treatments applied to certain series from external databases when necessary. Finally, it presents particular country-cases that required ad-hoc adjustments.

3.1 Data cleaning procedures for the INDSTAT Revision 3 Database

The ten raw unconsolidated series from INDSTAT primarily face three major issues[3]. First, some series contain spurious zeros that do not reflect the actual absence of activity but rather missing values that have been coded as zeros by the data providers (Pahl and M. P. Timmer, 2020). Second, some series contain duplicates, meaning that the same value is reported several times for a given country-subsector combination across different years. We address these cases differently depending on the type of variable.

[3]As a reminder, employment is expressed either in terms of the number of employees or the number of persons engaged, while value-added and output may each be reported at basic prices, producers' prices, factor cost, or using unknown valuation methods. The initial stage of the analysis thus contains 10 variables. For more details, please refer to Section 2.1 and Table 2

Table 4: Summary of the characteristics of each dataset used to construct the STiM database

Database	Aggregation level	ISIC	NVA reporting	Out. reporting	Emp. reporting	LCU/Dollars
INDSTAT	Disaggregated	Rev 3.1	Mix	Mix	Both	Current LCU
OECD STAN	Disaggregated	Rev 4	Basic p.	Basic p.	Both	Current LCU
OECD TiVA	Disaggregated	Rev 4	Basic p.	Basic p.	—	Current Dollars^{PWT}
OECD TiM	Disaggregated	Rev 4	—	—	Engaged	—
EU KLEMS	Disaggregated	Rev 4	Basic p.	Basic p.	Employees	Current LCU
ECLAC BADECON	Disaggregated	Rev 4	Unknown	Unknown	Unknown	Current Dollars^{PWT}
GGDC WIOD	Disaggregated	Rev 4	Basic p.	Basic p.	—	Current Dollars^{OG}
GGDC H-WIOD	Disaggregated	Rev 3.1	Producers p.	Producers p.	—	Current Dollars^{PWT}
APO	Disaggregated	Rev 4	Basic p.	—	Engaged	Current LCU
10SD/ASD/ETD	Aggregated	Rev 3.1	Basic p.	—	Engaged	Current LCU
ETD-TE	Aggregated	Rev 4	Basic p.	—	Engaged	Current LCU
WDI (World Bank)	Aggregated	Mix	Unknown	—	—	Current LCU

Note: This table reports the main characteristics of each source used to build the STiM database. The two columns 'NVA reporting' and 'Out. reporting' indicate whether value added and output are reported at basic prices, at factor cost, at producer prices, or as a mix of these different valuation methods. The column 'Employment reporting' indicates whether employment is reported as the number of employees, the number of persons engaged, or both. Note that if coverage is the same, we prefer basic prices over other valuation methods and employment reported as the number of employees over persons engaged. The last column indicates whether monetary variables are natively reported in local currency units (LCU) or in current US dollars. In the latter case, we specify whether the conversion relies on the Penn World Table (PWT). If 'OG' is specified, it means we have used original exchange rates (i.e., those that have been used to provide dollar estimates) to convert from dollars to LCU.

3.1.1 Detecting spurious zeros and dropping negative values

To begin with, we set all negative values to missing, as they do not make sense in the context of employment, value added, nor output. As shown in Table 2, these cases are confined to the unconsolidated value-added variables. We then follow Pahl and M. P. Timmer (2020) to detect zeros that are likely to correspond to missing values. First, we consider a zero spurious if it appears between two positive observations. Second, if a zero is reported in one of the three main variables of interest while at least one of the other two records a positive value, we also treat it as spurious. Third, if a zero is followed by a positive value such that the sub-sector suddenly accounts for more than 5% of total manufacturing, this zero is likely to reflect a missing value rather than a true observation. Similarly, a zero is treated as missing if it follows a year in which the sub-sector accounted for more than 5% of total manufacturing, since such an abrupt disappearance is much unlikely. Finally, we also treat zeros as missing when they occur between two missing values.

3.1.2 Dealing with duplicates

Some statistical offices in certain countries tend to report identical observations for the same sub-sectors over several consecutive years. While this may reflect a lack of updates and should therefore be treated as missing in most cases, we also consider that in some sub-sectors this phenomenon might be rational when reporting employment. Accordingly, we treat duplicates differently depending on the variable. If duplicates appear in employment for no more than two consecutive years within a given sub-sector, we retain them only if the total sum of manufacturing employment changes across these years. In this case, we assume that the duplicates reflect a temporary stagnation in that sub-sector. If duplicates in employment extend beyond two consecutive years, or if the total sum of manufacturing employment does not change, we consider only the earliest year as valid, with all subsequent values set to missing. For unconsolidated value-added and output series, the procedure is more straightforward as we do not tolerate any duplicates for any country-sub-sector combination. As with employment, we thus retain only the earliest year and set all subsequent duplicates to missing.

3.2 Data cleaning procedures and exchanges rates for the external databases

3.2.1 Data cleaning procedures

We apply minimal cleaning procedures to the external databases, as they have already been subjected to substantial harmonisation efforts by their respective providers. Furthermore, we seek to avoid compromising the internal consistency of the series we retrieve. Nonetheless, to ensure consistency with INDSTAT, we first remove all negative values from the value-added and output variables for each dataset incorporated. We also align the country coverage with that of UNIDO, restrict the variables to employ-

ment, value-added, and output, and limit the sectoral scope exclusively to manufacturing. Finally, all the variables are rescaled to ensure comparability, and the sample is restricted to the period up to 2019 to maintain consistency with INDSTAT. These steps are performed on every dataset included in the STiM database (i.e., Table 4). Note that for the WDI database from the World Bank, we also drop observations whenever manufacturing value added equals zero, which was the case for Brazil between 1963 and 1978 and for the Democratic Republic of Congo in 1991.

3.2.2 Exchange rates and conversions to LCU

Given we intend to rely on growth rates from external databases to complement INDSTAT, it is crucial to ensure that all series are expressed in the same currency unit as the latter. In Section 2.1, we mentioned that series were retrieved in local currency units (LCU) without further explanation. The rationale for this choice lies in several inconsistencies we identified in USD values from UNIDO arising from erroneous exchange rate estimates. Indeed, UNIDO initially provides series in both local currency units (LCU) and US dollars (USD), with the user able to choose which one to work with. As such, this allows anyone to derive the implicit exchange rates used for the conversion and compare them to other sources. When doing so, we identify a strong correlation between the exchange rate from UNIDO and the one from the Penn World Table (r=95). However, for a small subset of observations, differences with PWT were non-negligible and unrealistic. While most observations are clustered around 1 when computing a simple ratio between the two sources (i.e., no differences between the two exchange rate sources), extreme differences arise in the lower and upper 1% tails of the distribution (0.17 and 2.75×10^9). This implies that the bias affects both directions, with UNIDO exchange rates being either 0.17 times as large as those from PWT in the lower tail or about 2.75 billion times larger in the upper tail.

There are 110 country-year observations in this case, with seven countries affected, namely Ghana, Iraq, Lebanon, Myanmar, Nicaragua, Suriname, and Venezuela. Assuming that each country has observations for its 23 sub-sectors, this could lead to a potential bias affecting up to 1,320 observations. When repeating this exercise with the exchange rate used by the World Bank for calculating current total GDP, we found a similar pattern, which additionally identifies two other countries, Moldova and Syria. The case of Lebanon's exchange rate in 1998 and from 2007 to 2019 is quite illustrative of the kind of inconsistencies we observe. Following INDSTAT, the Lebanese pound remains stable over these years at an implausible level of one Lebanese pound per US dollar, while both PWT and the World Bank state that, on average, one US dollar equalled 1,507 Lebanese pounds. For this reason, we decide relying only on local currency units (LCU) series in INDSTAT, requiring therefore all external databases to also be expressed in LCU.

While most of them come in LCU, two databases are natively expressed in US dollars and thus require conversion. When possible, we use the same exchange rates as those used by the respective providers to recover original values. This is possible in the case

of the input–output table from the GGDC. Otherwise, when these are not stated or not available, we rely on exchange rates from the Penn World Table. Table 4 summarises whether the series from each external database were natively expressed in LCU or USD, and which exchange rates were used when necessary.

3.3 Specific country cases

One of the main advantages of INDSTAT is its extensive country coverage, which includes many developing economies that are often overlooked, as it relies on surveys conducted by national statistical offices. Interestingly, it also includes countries that previously existed but no longer do. We decide to treat these countries as separate entities when they split or reunify, leaving the choice to the user to reconstruct the series as desired. This means that we often assign non-existing country codes and names for these cases to distinguish them from current countries. For example, the former Federal Republic of Germany (West Germany) shared the same code as reunified Germany (DEU), which led us to assign a new country code for the former. The same reasoning applies to Sudan to distinguish pre-secession Sudan from the current Sudan after South Sudan's independence in 2011 (for which the country code would otherwise remain the same), or to Ethiopia and Eritrea prior to their separation in 1993. Table 5 summarises the specific country codes we assign to these cases.

Table 5: Special country codes

Country name	Assigned code
Czechoslovakia	CSK
Ethiopia and Eritrea	F-ETH
Germany, Fed Rep	FDR
Germany, Dem Rep	DDR
USSR	SUN
Yugoslavia	YUG
Serbia and Montenegro	SCG
Former Sudan	F-SDN

3.3.1 Serbia and Montenegro

Serbia and Montenegro were successively part of Yugoslavia until its dissolution in the early 1990s, then part of the Federal Republic of Yugoslavia (FRY) from 1992 to 2003 (a partially recognised state), then part of the State Union of Serbia and Montenegro (SCG) from 2003 to 2006, until the independence of Montenegro in 2006, which resulted in two sovereign states: Serbia (SRB) and Montenegro (MNG).

The INDSTAT database provides series for Yugoslavia from 1963 to 1989, i.e. three years before Yugoslavia's official dissolution. Subsequently, new series start under the name 'Serbia and Montenegro' from 1991 (corresponding to the FRY period) and extend until 2001 (employment) or 2002 (value added and output). In this latter year, series for

Serbia as a separate territorial entity also begin, while the series for Montenegro only starts in 2011. Without more details on the construction of these series, we decide to exclude the year 2002 from the 'Serbia and Montenegro' series to avoid overlapping with Serbia's own series in 2002 and thereby potentially double counting. Accordingly, we retain the 'Serbia and Montenegro' series from 1991 to 2001 and keep Serbia's series unchanged from 2002 onwards.

4 Aggregation into 12 manufacturing sub-sectors

In order to link the different data sources and address the aforementioned issues of combined activities, we had no choice but to aggregate the data into broader manufacturing sub-sectors. We proceed in two steps. First, we examined the 134 combinations of activities initially reported in INDSTAT and grouped sub-sectors according to the most frequent combinations observed. As discussed previously, these were mostly attributable to classification changes from ISIC Rev. 2 to ISIC Rev. 3 in 1989, as well as to unusual reporting practices. This procedure resulted in an aggregation into 12 manufacturing sub-sectors, identical to those used by Pahl and M. P. Timmer (2020) in their seminal paper. With respect to INDSTAT, observations corresponding to combinations that did not fit within this aggregation were dropped from the analysis. However, they represented less than 2% of total observations across each of the ten unconsolidated variables, which constitutes a negligible loss. The main trade-off of such aggregation is the reduction of cross sub-sector variation that could have been exploited by researchers, as the number of sub-sectors is reduced from 23 to 12. At the same time, this ensures a much higher degree of consistency and avoids comparing sub-sectors across years for which composition was changing.

Second, moving from ISIC Rev. 3.1 to ISIC Rev. 4 while remaining at the two-digit level inevitably leads to some misallocations. Mismatches between the two classifications necessarily arise due to changes at the three- and four-digit levels, even after several attempts to aggregate the initial 23 sectors into 12, 10, or 8 sub-sectors. Given that changes between ISIC Rev. 3.1 and ISIC Rev. 4 are too substantial and that no optimal solution exists, we decided to follow the correspondence applied by the OECD in their STAN database Horvát and Webb (2020, p.44). Although not perfect, this solution appeared to us to be the most reasonable given that their database have been used several times in the literature. Importantly, while their correspondence departs from the 23 initial sub-sectors, it remains compatible with our aggregation into 12 sub-sectors. This allows us to get a full correspondence between the classification that will be used in the STiM database (i.e., the new 12 sub-sectors) with ISIC Rev. 3, ISIC Rev. 3.1, ISIC Rev. 4 and NACE 2. Details are provided in Table 6.

When applying this classification to all data sources, it results to the loss of additional information beyond those already dropped because of an incompatibility between the current combination activity and the new classification. Indeed, the construction of the 12 aggregated sub-sectors logically requires complete information on all underlying

Table 6: Aggregation into 12 manufacturing sub-sectors

STiM, V1	**ISIC, Rev. 3.1**	**ISIC, Rev. 3**	**ISIC, Rev. 4**	**NACE 2**
1	15, 16	15, 16	10+11, 12	10+11, 12
2	17, 18, 19	17, 18, 19	13, 14, 15	13, 14, 15
3	20	20	16	16
4	21, 22	21, 22	17, 18	17, 18
5	23	23	19	19
6	24	24	20+21	20+21
7	25	25	22	22
8	26	26	23	23
9	27, 28	27, 28	24, 25	24, 25
10	29, 30, 31, 32, 33	29, 30, 31, 32, 33	28, 26, 27, 26, 26	28, 26, 27, 26, 26
11	34, 35	34, 35	29, 30	29, 30
12	36, 37	36	31+32+33	31+32+33

Note: This correspondence table is indirectly based on Pahl and M. P. Timmer (2020) and Horvát and Webb (2020) as discussed in the core of the paper. Labels for each sub-sector are provided in Table 7.

components. For instance, when creating the new sub-sector 'Food, beverages and tobacco products', if an observation is missing for either the manufacturing of food and beverages or the tobacco industry, then the entire aggregated observation for this aggregate sub-sector is not retained. This constraint ensures that all industrial activities are always consistently comparable across years. To complement the database we also link each of the 12 sub-sectors to their technological intensity following the OECD taxonomy (OECD, 2003) and Vu et al. (2021). This allows anyone to make use of the database for analyses related to manufacturing sophistication and technological upgrading. The correspondence between each sub-sector and its technological intensity is provided in Table 7, where we also attach reconstructed labels for the 12 sub-sectors. For the sake of flexibility and transparency, we also highlight that the replication files have been designed to allow users to easily adapt the re-aggregation process if needed. Moreover, for tractability purposes, two tag variables are created to link each new sub-sector to its original components in ISIC Rev. 3.1 and ISIC Rev. 4, respectively. These variables are included in the *detailled version* of the STiM database.

5 Linear interpolation

After aggregating all datasets into 12 sub-sectors, we perform a careful linear interpolation at the country, sub-sector, and variable levels for the INDSTAT Database[4]. This interpolation is applied within each unconsolidated series to preserve the specific levels associated with each valuation method, but it occurs *only* after the aggregation into twelve sub-sectors. This sequencing ensures that the interpolation relies on values that correspond to the correct industrial activity as defined by the panel, rather than on

[4]We don't apply this linear interpolation to the external datasets as they already have their own cleaning, harmonisation and inputting procedure.

Table 7: New aggregated labels

STiM, V1	New label	OECD Taxonomy
1	Food, beverages and tobacco products	Low-tech industries
2	Textiles, wearing apparel, leather products, fur	Low-tech industries
3	Wood products (excl. furniture)	Low-tech industries
4	Paper products, printing and publishing	Low-tech industries
5	Coke, refined petroleum products, nuclear fuel	Med-tech industries
6	Chemicals and chemical products	High-tech industries
7	Rubber and plastics products	Med-tech industries
8	Non-metallic mineral products	Med-tech industries
9	Basic and fabricated metal products	Med-tech industries
10	Machinery, equipment and electronic products	High-tech industries
11	Transport equipment	High-tech industries
12	Other manufacturing and recycling	High-tech industries

Note: The correspondence between each sub-sector and its technological intensity is derived from OECD (2003) and adapted following Vu et al. (2021). The STiM code number 12 — Other manufacturing and recycling — can be classified as either medium-tech or high-tech industries, since the sub-sectors it aggregates are typically assigned to both categories. To balance the classification (with four sub-sectors in each technological category), we assign it to high-tech industries.

one of the 134 combinations discussed earlier. However, we restrict interpolation to gaps of up to five consecutive years in a given series to avoid adding noise to the final estimates.

At this stage of the procedure, it applies to 4.2% of all employment observations, 2.4% of value-added observations, and 2.2% of output observations. For tractability purposes, a tag variable is created to identify which values have been interpolated in the additional *detailed version* of the STiM database.

6 The chain-linking procedure

In short, to ensure internal and intertemporal consistency within each country and across the three main variables of interest (employment, value added, and output), we implement a chain-linking procedure based on the growth rates of each series. This procedure, also used by Pahl and M. P. Timmer (2020), is standard in the literature on national accounts (Kruse et al., 2022). The idea is to establish an internal reference year common to all final variables for each country. Starting from this country-specific baseline, we reconstruct the consolidated series by extrapolating both backward and forward using the growth rates of the underlying unconsolidated series. In other words, this method assumes that the growth rates observed in each unconsolidated series correctly reflect the actual dynamics of the corresponding activity, such that combining these growth rates from a common reference year produces consistent series. The resulting consolidated series are internally consistent, since they are all derived from the same baseline and growth rates, and 'intertemporally' reliable, since no breaks in valuation methods occur once fixed by the reference year. Ultimately, to ensure

international comparability, all countries should share the same valuation method in the reference year (across the three main variable), which is the objective of our procedure, although this may not always be achievable given differences in national statistical office practices. This section aims at providing a detailed description of the chain-linking procedure.

6.1 Choosing the reference year

The first and crucial step of the chain-linking procedure involves determining a reference year that is specific to each country. This year serves as the baseline from which all other years are extrapolated using growth rates from the combined unconsolidated series. However, to ensure internal and intertemporal consistency, the reference year must be the same across the three main variables of interest within each country. Since we also aim to maximise both the length of the final series and the international comparability of the estimates, we design an algorithm to select the optimal reference year for each country.

- Step 1. Identifying the longest streak. We first tag the longest sequence of consecutive years during which each of the three main variables (employment, value added, and output) is available and covers at least one sub-sector. We then compute a score by summing the three variables, which allows us to identify the longest possible streak of coverage for each country.

- Step 2. Prioritising valuation methods. We then establish, ex ante, a preferred hierarchy of valuation methods for each variable. As recommended by the SNA (International Monetary Fund, 2009; International Monetary Fund, 2025), we give preference to employment series measured as the number of employees over those measured as the number of persons engaged. For value added and output, we prefer series expressed at basic prices over those at producer prices, and producer prices over factor prices. If none of these preferred methods are available, we fall back on the classification reported as 'unknown'.

- Step 3. Counting available observations. For each year, we compute the total number of non-missing observations across our 10 unconsolidated series that covers three main variables. Since we have 12 sub-sectors, the maximum possible count for one year is 36. Note that we ensure to lock final configurations when doing this count as the interpolation have created, sometimes, some overlap between two different reportings methods from the same variable.

- Step 4. Selecting the reference year. When having these pieces of information, we can proceed to the final step which consists in selecting the reference year. For each country, the reference year is chosen in the following order of priority:

1. The year must belong to the longest streak of consecutive years identified in the very first step. This avoids selecting isolated years with very limited surrounding

data, which is particularly important since the extrapolation of values is highly sensitive to missing observations in the growth-rate series.

2. Among these years, we first select those when employment is reported as the number of employees. If none exist, we retain years when employment is reported as the number of persons engaged.

3. Within this set, we further restrict to years when value added is reported according to our preferred order of valuation methods (basic prices, then producer prices, then factor prices, then unknown classification).

4. Still retaining these years, we apply the same preference order to output. If possible, output should be valued using the same method chosen for value added to ensure consistency. If this is not possible, we relax this condition and select the second preferred method for output, and so on (basic prices, then producer prices, then factor prices, then unknown classification).

5. Among the remaining years, we choose the one with the highest count of non-missing observations across the three variables.

6. Lastly, if multiple years still satisfy all these conditions, we select the most recent one as the reference year.

Note that, for tractability purposes, one tag variable is created to identify the reference year selected for each country, along with three additional tag variables that record the valuation method used for each of the three main variables in that year. These variables are particularly useful for researchers who wish to avoid conducting cross-country analyses on value added, output, or employment when the underlying levels are not based on the same valuation method. These tag variables are included in the *detailed version* of the STiM database.

6.2 Combining the growth rates from all available disaggregated sources

Once a unique reference year is selected for each country, we proceed to the second step of the chain-linking procedure, which involves combining the growth rates of the three main variables of interest. At this stage, external 'disaggregated' sources are integrated into the process thanks to the aggregation discussed in the previous section. We thus construct three final growth-rate series — one for employment, one for value added, and one for output. Each combined growth-rate series encompasses all available growth rates from every dataset and valuation method. Importantly, these combined growth-rate series are built from pre-computed growth rates within each unconsolidated source. In other words, for INDSTAT, we first compute growth rates within each valuation method and then combine them, rather than merging raw values across valuation methods to derive growth rates. This approach consistent with international recommendations ensures that each rate reflects a consistent definition of the underlying variable.

For instance, the final employment growth-rate series used to reconstruct employment levels is constructed according to a deterministic order of priority. If available, the series relies on employment growth rates reported by UNIDO, with preference given to measures expressed as the number of employees over the number of persons engaged. If these are not available, growth rates are drawn from external sources following a predefined hierarchy. The same logic applies to value-added and output growth-rate series, where priorities depend on the preferred valuation methods. When multiple series rely on the same valuation method, INDSTAT is systematically prioritised over external sources, as it acts as the primary data source for the STiM database. Table 8 summarises the full set of priority rules used to combine growth rates across sources, measurements, and valuation methods. Note that we usually prefer OECD STAN as the second-best source because the database has been specifically designed to provide detailed, harmonised industry-level data. Given that EU-KLEMS involves substantial aggregation and is rather limited in coverage, we also usually prefer other alternatives to it, notably TiVA and datasets from the Groningen Growth and Development Center (GGDC). As for APO, it is often the last source to be used, as there is still some uncertainty regarding its re-aggregation into 12 sub-sectors, since the correspondence table to ISIC Rev.4 is incomplete. Growth rates from aggregated sources are added to the combined growth rates series at a later stage; more details about this step are provided in the Section .

Note that, for tractability purposes, three tag variables are created to identify the source of the growth rate for each of the three main variables in each country-year. It allows researchers to easily track the source of the final estimated values. These tag variables are included in the *detailed version* of the STiM database.

6.3 Assuming constant labour-productivity to bridge minor gaps in growth rates combined series

Whenever a change in reporting methods occurs, it mechanically generates a break in the combined growth-rate series of a single variable (e.g., the combined growth rate of all value-added measures). This happens because growth rates are computed separately within each unconsolidated series before being combined. This approach is the only way to ensure that each rate effectively captures a consistent definition of the underlying variable. By contrast, combining raw values first and then computing growth rates would imply calculating changes across reporting methods, thereby conflating methodological breaks with genuine economic dynamics. Although this procedure follows most international recommendations, it has one drawback in our setting. Since there are two to four possible reporting methods for a single final measure, each change in valuation creates a one-year gap in the final growth-rate series of the variable concerned. As a result, when a country frequently changes its reporting methods over time, the final growth-rate series will systematically contain (at least) one-year gaps at each valuation break because each segment originates from a distinct unconsolidated source. To address this issue and prevent the extrapolation process

Table 8: Priority rules for combining growth rates across sources and valuation methods

	Employment growth rates		Value-added growth rates		Output growth rates	
Priority	Source	Measurement	Source	Valuation	Source	Valuation
1	UNIDO	Nb. employees	UNIDO	Basic prices	UNIDO	Basic prices
2	OECD STAN	Nb. employees	OECD STAN	Basic prices	OECD STAN	Basic prices
3	EU-KLEMS	Nb. employees	OECD TiVA	Basic prices	OECD TiVA	Basic prices
4	APO	Nb. employees	GGDC (WIOD)	Basic prices	GGDC (WIOD)	Basic prices
5	UNIDO	Persons engaged	EU-KLEMS	Basic prices	EU-KLEMS	Basic prices
6	OECD STAN	Persons engaged	APO	Basic prices	UNIDO	Producer prices
7	OECD TiM	Persons engaged	UNIDO	Producer prices	GGDC (H-WIOD)	Producer prices
8	CEPAL Badecon	Unknown	GGDC (H-WIOD)	Producer prices	UNIDO	Factor cost
9	–	–	UNIDO	Factor cost	UNIDO	Unknown
10	–	–	UNIDO	Unknown	CEPAL Badecon	Unknown
11	–	–	CEPAL Badecon	Unknown	–	–

Notes: Priority levels indicate the deterministic order in which growth rates are selected for each country–year. Within each block, the first available growth rate according to this hierarchy is retained. UNIDO acts as the primary source whenever comparable definitions are available. For aggregated datasets, GGDC series are preferred to WDI indicators.

from stopping at each missing observation, we assume constant labour productivity to bridge these minor gaps in the growth-rate series. We make this assumption in line with Pahl and M. P. Timmer (2020) and apply it only when two growth-rate series can be bridged. In other words, if two value-added growth-rate series corresponding to different reporting methods are separated by a one-year gap in the final combined value-added growth rate series, we assume that the missing growth rate for that year equals the growth rate of employment in the same year. The same logic applies to output series, which rely on the growth rate of employment to fill one-year gaps (i.e., assuming constant labour productivity per worker for output). Alternatively, if a one-year gap occurs in the employment growth-rate series, we assume that the missing growth rate equals the growth rate of value added, without relying on output. This choice reflects the closer conceptual relationship between employment and value added.

In the current version of the STiM database, we tolerate up to three missing values between two growth-rate series. This means that we assume, at most, three consecutive years of constant labour productivity within the same series to bridge such gaps. This assumption is introduced to prevent the extrapolation process from being interrupted too frequently, particularly for countries that frequently change their reporting methods or take time to adapt their reporting systems. If more than three consecutive years are missing, the constant labour productivity assumption is not applied, meaning that the remaining part of the series is lost. In addition, it also implies that if a country changes in the same year the valuation methods for value-added/output and employment, no bridging is possible since no alternative variable is available.

As an example, this was the case for Paraguay in 2010. The national statistical institute switched from measuring employment as the number of employees to the number of persons engaged, while simultaneously changing the valuation of value added and output from producers' prices to an unspecified method. As a result, no bridging is possible in this case, since the growth rate of employment cannot be used to fill the gap in value-added or output growth rates, and vice versa. Importantly, the labour productivity computed from 2010 onward — based on the new definitions of both employment and valuation methods — cannot be used to link pre-2010 and post-2010 segments, as doing so would artificially merge two incompatible measurement levels and bias the resulting chained series. We therefore cannot recover values after 2009 in the absence of external datasets whose growth rates could be used to fill the gap for at least that year. If such growth rates were available, it would then be possible to use those from INDSTAT in subsequent years.

Although the loss of some series separated by more than three years cannot be entirely avoided, we recall that we mitigate this issue by selecting the reference year within the longest continuous non-missing streak of the series. This approach prevents us from relying solely on valuation-method preferences, which could otherwise result in very short final series when the preferred valuation is reported for only a few years (e.g., Eswatini's case). Overall, in our final estimates, assumptions of constant labour

productivity corresponding to only 0.7% of observations in the final employment growth-rate series, 0.9% in the final value-added growth-rate series, and 1% in the final output growth-rate series. Nonetheless, for transparency, three tag variables are created to identify, for each final series, the values estimated under the assumption of constant labour productivity growth. These tag variables are included in the *detailed version* of the STiM database.

6.4 Integrating aggregated datasets to combined growth rates

After having combined growth rates from all available disaggregated sources, we integrate growth rates from aggregated datasets (e.g., WDI, GGDC) into the process. While this hypothesis might seem quite strong, we restrict it to cases where two growth-rate streaks are separated such that we would lose part of the streak if we did not combine them. As discussed in the next section, the chain-linking procedure relies on growth rates from the reference year to reconstruct the series, such that any interruption in the series that could not be resolved by linear interpolation (either because the gap is too long or because there has been a change in valuation methods in INDSTAT), nor by the CLP hypothesis, would lead to a loss of part of the series. For this reason, we only fill such gaps with growth rates from aggregated sources, thus assuming for a limited number of years that the sub-sectors follow the same growth rate as the whole manufacturing industry. In case of multiple sources between GGDC and the World Bank, we always prefer GGDC series, as they are harmonised and follow SNA standards. Note that, for tractability, tag variables are created to identify these cases. These tag variables are included in the *detailed version* of the STiM database.

6.5 Backward and forward extrapolation

The final step of the chain-linking procedure involves reconstructing the consolidated series by extrapolating both backward and forward from the reference year using the combined growth-rate series. The reference-year value (X_{t_0}) for each of the three consolidated series is always taken from the INDSTAT database. The extrapolation is performed as follows, where g_t denotes the growth rate stemming from the combined employment, value-added, and output growth-rate series:

$$X_t = \begin{cases} X_{t-1} \times (1 + g_t), & \text{for } t > t_0 \quad \text{(forward extrapolation)} \\ X_{t+1} \div (1 + g_{t+1}), & \text{for } t < t_0 \quad \text{(backward extrapolation)} \end{cases} \tag{1}$$

6.6 Additional reference years to recover extra observations

Once this first extrapolation is complete, we perform additional extrapolations by allowing a given country to have several reference years if the initial streak is interrupted by a series of missing values. However, each new reference year must be consistent with

the valuation method of the first one in order to preserve internal and intertemporal consistency. To be more precise, we adopt as many additional reference years as there are complete breaks common to all three consolidated variables in a given country. Recall that this procedure is performed after an initial attempt to avoid these gaps through the CLP hypothesis and the initial five-year interpolation. Each new reference year is locked to the same configuration (employment concept and price/valuation) as the original reference year and is chosen once per complete post-break streak. Within that streak, we apply the same selection filters as for the first reference year. We then chain-link starting from this new anchor using growth rates within that streak.

If, in a given streak, the exact configuration is not simultaneously available for all three variables, we nonetheless define a single common reference year for the streak, anchoring it to the largest admissible subset according to the following priority order. Suppose two out of three variables share the same configuration as the original reference year. In that case, we prefer to recover (i) employment and value added if available, followed by (ii) employment and output, and lastly (iii) output and value added. If no pair is available, we allow single-variable anchors in the following preference order: (i) employment, followed by (ii) value added, and then (iii) output. In all cases, the reference year is unique and applies simultaneously to the three variables within the same streak, ensuring intertemporal consistency. In all cases, the reference year is unique and applies simultaneously to the three variables within the same streak, ensuring internal consistency. This is crucial to avoid distortions in the series, as failing to respect this condition would misalign levels and distort key ratios — notably labour productivity when measured as value added per worker for example. It is therefore impossible to assign a reference year to only one variable, while the series for another variable continue from the previous chain-linking. Put differently, additional reference years are introduced only when a total break occurs — namely, a common discontinuity in employment, value added, and output that cannot be bridged by interpolation or the CLP hypothesis.

The case of Tunisia illustrates why we aim to implement this procedure. In Tunisia, there is a common break across employment, value added, and output from 1982 to 1988 that cannot be bridged by interpolation or CLP. As such, we initially end up with two separate streaks: 1963-1981 and 1989-2019. Given the algorithm used to select the reference year, the first anchor is set in the longest streak (1989-2019) and corresponds to 2002 which maximise the coverage and valuation methods. In that year, value added and output are reported at basic prices, while employment is measured as the number of persons engaged. To avoid losing the whole 1963-1981 series, we introduce a second reference year for the earlier streak, locking it to the same configuration as in 2002 — producers' prices and number of employees. Since several matching years are available, we then choose among these years the one with the highest count of non-missing observations and the most recent one in this streak, which is 1981. This procedure recovers the whole 1963-1981 block while preserving internal consistency (the same valuation/definitions within each anchor) and intertemporal consistency with the most recent streak. It thus extends the usable series without mixing configurations.

7 Deflating procedure

Natively, INDSTAT reports value added and output in nominal terms. To disentangle price from quantity movements, we follow Haraguchi and Amann (2023) recommendations. We therefore exploit the Index of Industrial Production (IIP) embedded in the INDSTAT database, which measures real production growth by manufacturing division relative to the 2015 benchmark. To be more precise, let us denote nominal output O for country i, sub-sector s, and year t as the following product between prices and quantity.

$$O_{ist} = P_{ist} \times Q_{ist} \tag{2}$$

If we consider that $\tilde{O}_{ist}$ denotes constant-price output valued at base-year prices P_{isb} for a given baseline b, we can express the IIP as follows:

$$\text{IIP}_{ist} = \frac{\tilde{O}_{ist}}{O_{isb}} = \frac{P_{isb} \times Q_{ist}}{P_{isb} \times Q_{isb}} = \frac{Q_{ist}}{Q_{isb}}. \tag{3}$$

Hence, the relative output price is identified from nominal output and the IIP:

$$\frac{P_{ist}}{P_{isb}} = \frac{O_{ist}/O_{isb}}{\text{IIP}_{ist}} \tag{4}$$

As such, retrieving the embedded price index (PI) at a base year b can be done in the following way:

$$\text{PI}_{ist} = 100 \times \frac{O_{ist}/\text{IIP}_{ist}}{O_{isb}/\text{IIP}_{isb}} \tag{5}$$

Note that PI_{ist} is computed before any ex-post cleaning or re-aggregation is done so that the decomposition respects the original accounting identity. In other words, PI_{ist} is retrieved using the initial consolidated raw output series, which mixes different valuation methods. While this preserves coherence with the initial framework, the resulting index is later applied to the cleaned nominal output and value-added series.

In this single-deflation setting, we set $b = 2015$ for all countries and sub-sectors, as it is the native reference year of the IIP index. Since the IIP is initially reported for the 22 original ISIC manufacturing sub-sectors, the next step implies aggregating these indices to match the STiM classification into 12 broader sub-sectors. This aggregation is carried out by computing a weighted average of the individual price indices, where the weights correspond to the nominal output of each original sub-sector in each year. As such, if we note S one of the new STiM sub-sectors composed of a set of ISIC sub-sectors j, the price index for a STiM sub-sector S in year t is given by:

$$\text{PI}_{S,t} = \frac{\sum_{j \in S} O_{j,t} \times \text{PI}_{j,t}}{\sum_{j \in S} O_{j,t}} \tag{6}$$

This rule is applied for S1 (15–16), S2 (17–19), S4 (21–22), S9 (27–28), S10 (29–33), and S11 (34–35). For S12 (36–37), a weighted average cannot be performed since ISIC 37 is not covered by the IIP. As a result, we apply the price index of ISIC 36 to the entire

group. The remaining sectors (S3, S5, S6, S7, and S8) correspond directly to a single ISIC sub-sector, so their price index is directly inherited from the corresponding ISIC sub-sector. More details about the aggregation are provided in the previous section (i.e., Table 4).

However, a complete weighted average is not always possible. To illustrate, consider the STiM sub-sector S2, which aggregates three underlying ISIC sub-sectors (17, 18, 19). In theory, the final price index for S2 would require the three corresponding price indices and the output for each of these sub-sectors. Yet, if only two price indices (or output values) out of the three required to aggregate correctly into S2 are available, we compute a partial weighted average over those two. If only one is available out of the three, we use that one as a proxy for the whole sector. In other words, when a complete weighted average cannot be computed, we assume that the missing sub-sectors j within S exhibit the same price dynamics as the available ones. If none is available, we use the next-best alternative source of deflator (see below).

Note that for tractability purposes, we provide a tag variable that documents whether the price index for a given sector is based on a full or partial weighted average. This tag is only available when the deflator used is derived from the IIP. The tag is only available in the *detailed version* of the STiM database.

As such, to obtain real value-added and constant output:

$$\mathrm{VA}_{ist}^{\mathrm{real}} = \frac{\mathrm{VA}_{ist}^{\mathrm{nom}}}{\mathrm{PI}_{ist}^{\mathrm{VA}}} \times 100 \quad ; \quad \mathrm{Y}_{ist}^{\mathrm{real}} = \frac{\mathrm{Y}_{ist}^{\mathrm{nom}}}{\mathrm{PI}_{ist}^{\mathrm{Y}}} \times 100 \tag{7}$$

8 Final database

At the end of the cleaning and chain-linking procedure, we obtain a final database covering 12 manufacturing sub-sectors across more than 145 countries from 1963 to 2019. The final STiM database contains 56,801 observations for value added, 58,574 for output, and 58,414 for employment (i.e., Table 9). Yet, it should be noted that all the cleaning, aggregation, and harmonisation procedures described in the previous sections lead to a loss of approximately half of the initial observations when compared with the original raw data at the 22-sector level (i.e., Table 2). After the chain-linking procedure, the final valuation structure of the STIM database shows that most observations are valued at producers' prices (41.62% and 47.91% of total observations for value added and output, respectively). These are followed by basic prices (23.55% and 19.48%) and factor cost valuations (28.18% and 18.46%). The remaining observations (6.65% and 14.16%) correspond to cases where the valuation method is unknown. In other words, despite the efforts to prefer basic prices for both value added and output, the most common valuation method in the final database is producers' prices. This is partly explained by our choice to restrict the initial reference year to the longest initial streak of observations in the original algorithm designed to select the optimal reference years. While these cross-country differences in valuation methods may weaken the

Table 9: Statistics regarding valuation methods in the STiM Database

	Value added		Output		Employment	
Configuration	N	%	N	%	N	%
Basic prices	13,375	23.55	11,409	19.48	—	—
Producers prices	23,639	41.62	28,061	47.91	—	—
Factor values	16,007	28.18	10,810	18.46	—	—
Unknown prices	3,780	6.65	8,294	14.16	—	—
Number of employees	—	—	—	—	56,911	97.43
Number of engaged	—	—	—	—	1,503	2.57
Total	**56,801**	**100.00**	**58,574**	**100.00**	**58,414**	**100.00**

international consistency of the database, any resulting bias is now country-specific and time-invariant thanks to the chain-linking that preserves initial level differences. Whatever its sign, such bias would therefore be theoretically captured by including country fixed effects in any econometric analyses. Accordingly, we strongly recommend that all empirical analyses using the STiM database include country dummies to account for level differences arising from valuation methods. Regarding employment, almost all observations refer to the number of employees (97.43%), while only 2.57% refer to the number of persons engaged.

Regarding the sources that contributes to the final database, the most important one is UNIDO although one would note a clear bias towards employment (92%), while covering half of the value added and output observations (50% and 55%, respectively). The second most important source is the OECD STAN database, which contributes to 19% of value added and 18% of output observations, but only 3% of employment. The third most important source is the OECD TiVA database, which contributes to 17% of value added and 17% of output observations. The remaining sources contribute less than 5% of the final database for each variable, and the contribution of aggregated data sources is rather limited, which is both reassuring and a consequence of restricting their use to filling gaps between disconnected growth-rate streaks (already existing). More details are provided in table 10.

9 Conclusion

This paper sets out to present the construction of the STiM database, which provides employment, value-added, and output data at the two-digit manufacturing sub-sectoral level for more than 145 countries from 1963 to 2019. The database is the result of combining several data sources and undertaking substantial harmonisation efforts aimed at ensuring internal, intertemporal, and international consistency. In addition, it aims linking several price deflators from multiple institutions to adjust nominal measures and produces consistent estimates of real value-added and real output. Monetary variables are so far expressed in local currency unit (LCU.) To date and to

Table 10: Distribution of growth-rate sources by variable (percentages)

Source	Employment	Value added	Output
UNIDO (INDSTAT)	92.41	50.33	55.28
OECD STAN	2.91	19.39	17.86
EU-KLEMS	0.71	3.04	3.29
OECD TiVA	–	16.60	17.21
OECD TiM	1.70	–	–
APO Database	0.32	2.27	–
ECLAC Badecon	0.13	0.01	0.01
GGDC (WIOD)	–	0.09	0.26
GGDC (H-WIOD)	–	4.79	6.09
GGDC ETD-10SD-AESD	1.73	0.98	–
GGDC ETD-TE	0.09	0.01	–
WDI (World Bank)	–	2.48	–
Total	100.00	100.00	100.00

our knowledge, this makes the Structural Transformation in Manufacturing database (STIM) the most comprehensive dataset for long-run analyses of structural change within manufacturing in both developed and developing economies.

Besides contributing to the existing literature by extending previous academic efforts to compile and harmonise macroeconomic statistics for a wide range of countries, the database also aims to promote further research avenues on how manufacturing has evolved over the last six decades. Moreover, it might be also used to explore how manufacturing can still be leveraged in our modern economies to foster inclusion, growth, and development.

Finally, we recall that the final database comes under two versions. The *ready-to-use version* contains the five core variables discussed in the paper (employment, nominal value added, real value added, nominal output, and real output), along with metadata identifying the country, year, and relevant manufacturing sub-sector. This release is also complemented by a *detailed version*, which only includes tag variables designed to enhance the transparency and tractability of the harmonisation and estimation procedures. The database and the replication package will be made freely available through the corresponding GitHub repository, enabling researchers to replicate our results, build upon our work, and contribute to the improvement of the database. This also serves to reaffirm that the current release is only the first iteration of the STiM database (V1.0), and we welcome feedback and suggestions for future updates and enhancements. Additionally, further data sources are planned to be integrated as they become available, thereby enhancing both the quality and coverage of the database.

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10 Appendix

Table A1: Manufacturing codes in the OECD STAN database

	Employees		**Engaged**		**N. Value Added**		**N. Output**	
	Obs.	%	Obs.	%	Obs.	%	Obs.	%
C10	628	2.00	718	1.87	808	1.94	761	2.00
C11	628	2.00	716	1.86	804	1.93	757	1.98
C12	719	2.29	887	2.31	976	2.34	896	2.35
C13	717	2.29	854	2.22	997	2.39	886	2.32
C14	717	2.29	848	2.21	997	2.39	886	2.32
C15	818	2.61	976	2.54	1030	2.47	918	2.41
C16	905	2.89	1093	2.84	1163	2.79	1079	2.83
C17	905	2.89	1093	2.84	1163	2.79	1079	2.83
C18	905	2.89	1086	2.82	1178	2.82	1094	2.87
C19	895	2.85	1135	2.95	1245	2.99	1109	2.91
C20	883	2.82	1078	2.80	1217	2.92	1086	2.85
C21	858	2.74	1046	2.72	1188	2.85	1057	2.77
C22	877	2.80	1044	2.71	1108	2.66	1029	2.70
C23	877	2.80	1044	2.71	1123	2.69	1064	2.79
C24	902	2.88	1090	2.83	1160	2.78	1096	2.87
C25	905	2.89	1093	2.84	1163	2.79	1099	2.88
C26	925	2.95	1078	2.80	1203	2.88	1144	3.00
C27	925	2.95	1070	2.78	1212	2.91	1153	3.02
C28	932	2.97	1151	2.99	1294	3.10	1158	3.04
C29	888	2.83	1050	2.73	1083	2.60	1044	2.74
C30	888	2.83	1046	2.72	1084	2.60	1045	2.74
C33	858	2.74	957	2.49	952	2.28	919	2.41
C10_11	725	2.31	909	2.36	1006	2.41	924	2.42
C13_14	835	2.66	1001	2.60	1057	2.53	940	2.46
C20_21	901	2.87	1142	2.97	1253	3.00	1117	2.93
C22_23	955	3.05	1195	3.11	1318	3.16	1182	3.10
C24_25	955	3.05	1248	3.25	1342	3.22	1200	3.15
C26_27	932	2.97	1158	3.01	1289	3.09	1153	3.02
C29_30	955	3.05	1248	3.25	1342	3.22	1200	3.15
C31_32	858	2.74	962	2.50	955	2.29	922	2.42
C10T12	955	3.05	1248	3.25	1342	3.22	1200	3.15
C13T15	956	3.05	1249	3.25	1342	3.22	1200	3.15
C16T18	955	3.05	1248	3.25	1342	3.22	1200	3.15
C19T23	903	2.88	1197	3.11	1289	3.09	1148	3.01
C26T28	955	3.05	1248	3.25	1341	3.22	1199	3.14
C31T33	955	3.05	1248	3.25	1342	3.22	1200	3.15
Total	31350	100.00	38454	100.00	41708	100.00	38144	100.00

Note: This table reports the number of non-missing observations by manufacturing sub-sector in the OECD STAN database over the period 1970-2019, restricted to data at the two-digit level. Simple codes correspond to single two-digit manufacturing industries. Codes with an underscore (e.g., C10_11) denote exact partial aggregates obtained by summing the corresponding two-digit activities, while those containing a 'T' represent contiguous range aggregates over the whole range (e.g., C10T12). The details of each code are provided in Table 3.

Table A2: Manufacturing codes in the OECD TiVA database

	Nominal Value Added		**Nominal Output**	
	Observations	%	Observations	%
C16	2000	4.35	2000	4.35
C19	2000	4.35	2000	4.35
C20	2000	4.35	2000	4.35
C21	2000	4.35	2000	4.35
C22	2000	4.35	2000	4.35
C23	2000	4.35	2000	4.35
C24	2000	4.35	2000	4.35
C25	2000	4.35	2000	4.35
C26	2000	4.35	2000	4.35
C27	2000	4.35	2000	4.35
C28	2000	4.35	2000	4.35
C29	2000	4.35	2000	4.35
C30	2000	4.35	2000	4.35
C17_18	2000	4.35	2000	4.35
C20_21	2000	4.35	2000	4.35
C24_25	2000	4.35	2000	4.35
C26_27	2000	4.35	2000	4.35
C29_30	2000	4.35	2000	4.35
C10T12	2000	4.35	2000	4.35
C13T15	2000	4.35	2000	4.35
C16T18	2000	4.35	2000	4.35
C19T23	2000	4.35	2000	4.35
C31T33	2000	4.35	2000	4.35
Total	46000	100.00	46000	100.00

Note: This table reports the number of non-missing observations by manufacturing sub-sector in the OECD TiVA database over the period 1995–2019, restricted to data at the two-digit level. Simple codes correspond to single two-digit manufacturing industries. Codes with an underscore (e.g., C10_11) denote exact partial aggregates obtained by summing the corresponding two-digit activities, while those containing a 'T' represent contiguous range aggregates over the whole range (e.g., C10T12). The details of each code are provided in Table 3.

Table A3: Manufacturing codes in the OECD TiM database

	Number of persons engaged	
	Observations	Percentage
C16	1525	4.35
C19	1525	4.35
C20	1525	4.35
C21	1525	4.35
C22	1525	4.35
C23	1525	4.35
C24	1525	4.35
C25	1525	4.35
C26	1525	4.35
C27	1525	4.35
C28	1525	4.35
C29	1525	4.35
C30	1525	4.35
C17_18	1525	4.35
C20_21	1525	4.35
C24_25	1525	4.35
C26_27	1525	4.35
C29_30	1525	4.35
C10T12	1525	4.35
C13T15	1525	4.35
C16T18	1525	4.35
C19T23	1525	4.35
C31T33	1525	4.35
Total	35075	100.00

Note: This table reports the number of non-missing observations by manufacturing sub-sector in the OECD TiM database over the period 1995–2019, restricted to data at the two-digit level. Simple codes correspond to single two-digit manufacturing industries. Codes with an underscore (e.g., C10_11) denote exact partial aggregates obtained by summing the corresponding two-digit activities, while those containing a 'T' represent contiguous range aggregates over the whole range (e.g., C10T12). The details of each code are provided in Table 3.

Table A4: Manufacturing codes in the EU KLEMS database

	Employees		**N. Value Added**		**N. Output**	
	Obs.	%	Obs.	%	Obs.	%
C19	692	6.51	667	6.34	667	6.29
C20	734	6.91	684	6.51	684	6.45
C21	628	5.91	623	5.93	623	5.88
C26	700	6.59	695	6.61	695	6.56
C27	706	6.64	706	6.72	706	6.66
C28	702	6.61	702	6.68	702	6.62
C10-C12	725	6.82	725	6.90	725	6.84
C13-C15	750	7.06	750	7.13	750	7.07
C16-C18	725	6.82	725	6.90	725	6.84
C20-C21	685	6.45	655	6.23	700	6.60
C22-C23	725	6.82	725	6.90	725	6.84
C24-C25	725	6.82	725	6.90	725	6.84
C26-C27	681	6.41	681	6.48	725	6.84
C29-C30	725	6.82	725	6.90	725	6.84
C31-C33	725	6.82	725	6.90	725	6.84
Total	10628	100.00	10513	100.00	10602	100.00

Note: This table reports the number of non-missing observations by manufacturing sub-sector in the EU KLEMS database over the period 1995–2019, restricted to data at the two-digit level. Simple codes correspond to single two-digit manufacturing industries. Codes with a hyphen (e.g., C10-C11) denote exact partial aggregates obtained by summing the corresponding two-digit activities. The details of each code are provided in Table 3.

Table A5: Manufacturing codes in the CEPAL BADECON database

	Employees		**Nominal Value Added**		**Nominal Output**	
	Observations	%	Observations	%	Observations	%
C10	55	4.53	45	4.40	45	4.36
C11	50	4.12	45	4.40	45	4.36
C12	30	2.47	21	2.05	22	2.13
C13	55	4.53	45	4.40	45	4.36
C14	50	4.12	40	3.91	40	3.88
C15	50	4.12	45	4.40	45	4.36
C16	55	4.53	45	4.40	45	4.36
C17	55	4.53	45	4.40	45	4.36
C18	52	4.28	42	4.11	42	4.07
C19	45	3.71	35	3.42	35	3.39
C20	55	4.53	45	4.40	45	4.36
C21	50	4.12	45	4.40	45	4.36
C22	55	4.53	45	4.40	45	4.36
C23	50	4.12	45	4.40	45	4.36
C24	55	4.53	45	4.40	45	4.36
C25	55	4.53	45	4.40	45	4.36
C26	51	4.20	42	4.11	42	4.07
C27	55	4.53	45	4.40	53	5.14
C28	50	4.12	45	4.40	45	4.36
C29	50	4.12	45	4.40	45	4.36
C30	51	4.20	42	4.11	42	4.07
C31	47	3.87	42	4.11	42	4.07
C32	50	4.12	45	4.40	45	4.36
C33	33	2.72	28	2.74	28	2.72
CXX	10	0.82	10	0.98	10	0.97
Total	1214	100.00	1022	100.00	1031	100.00

Note: This table reports the number of non-missing observations by manufacturing subsector in the CEPAL BADECON database over the period 2010–2019, restricted to data at the two-digit level. The details of each code are provided in Table 3.

Table A6: Manufacturing codes in the GGDC-WIOD database

	Nominal Value Added		**Nominal Output**	
	Observations	%	Observations	%
C16	645	5.27	645	5.27
C17	645	5.27	645	5.27
C18	645	5.27	645	5.27
C19	632	5.16	632	5.16
C20	645	5.27	645	5.27
C21	645	5.27	645	5.27
C22	645	5.27	645	5.27
C23	645	5.27	645	5.27
C24	645	5.27	645	5.27
C25	645	5.27	645	5.27
C26	645	5.27	645	5.27
C27	645	5.27	645	5.27
C28	645	5.27	645	5.27
C29	645	5.27	645	5.27
C30	645	5.27	645	5.27
C33	645	5.27	645	5.27
C10T12	645	5.27	645	5.27
C13T15	645	5.27	645	5.27
C31T32	645	5.27	645	5.27
Total	12242	100.00	12242	100.00

Note: This table reports the number of non-missing observations by manufacturing sub-sector in the GGDC-WIOD database over the period 2000-2014, restricted to data at the two-digit level. The details of each code are provided in Table 3.

Table A7: Manufacturing codes in the Historical GGDC-WIOD database

	Nominal Value Added		**Nominal Output**	
	Observations	%	Observations	%
D23	900	8.33	900	8.33
D24	900	8.33	900	8.33
D25	900	8.33	900	8.33
D26	900	8.33	900	8.33
D29	900	8.33	900	8.33
Dnec	900	8.33	900	8.33
D15t16	900	8.33	900	8.33
D17t19	900	8.33	900	8.33
D21t22	900	8.33	900	8.33
D27t28	900	8.33	900	8.33
D30t33	900	8.33	900	8.33
D34t35	900	8.33	900	8.33
Total	10800	100.00	10800	100.00

Note: This table reports the number of non-missing observations by manufacturing sub-sector in the Historical GGDC-WIOD database over the period 1965-2000, restricted to data at the two-digit level. The details of each code are provided in Table 1 given that codes follow ISIC 3.1. Revision.

Table A8: Manufacturing codes in the APO database

	Employees		**Nominal Value Added**	
	Observations	%	Observations	%
C16	257	7.73	695	8.47
C23	257	7.73	719	8.77
C24	257	7.73	694	8.46
C27	257	7.73	511	6.23
C25_26	226	6.79	298	3.63
C28_29	255	7.67	514	6.27
C30_31	239	7.19	522	6.36
C10T12	297	8.93	719	8.77
C13T15	253	7.61	719	8.77
C17T18	257	7.73	719	8.77
C19T22	257	7.73	719	8.77
C25T31	257	7.73	668	8.14
C32T33	257	7.73	705	8.60
Total	3326	100.00	8202	100.00

Note: This table reports the number of non-missing observations by manufacturing sub-sector in the Asian Productivity Organisation database over the period 1970-2019, restricted to data at the two-digit level. Codes with an underscore (e.g., C25_26) denote exact partial aggregates obtained by summing the corresponding two-digit activities, while those containing a 'T' represent contiguous range aggregates over the whole range — and therefore often correspond to broader groupings (e.g., C25T31). The details of each code are provided in Table 3.